

MUHAMMAD MAHWIZ KHALIL

+923330214897 | khalilmahwiz@gmail.com | LinkedIn | Karachi, Pakistan

SUMMARY

Python Engineer specializing in Deep learning systems and AI infrastructure. Proven track record of architecting and deploying NLP solutions with measurable business impact. Expert in building scalable distributed systems using frameworks and cloud technologies.

CORE TECHNICAL COMPETENCIES

Programming & Architecture: Python, C, SQL, System Design, Distributed Computing
Backend Engineering: Flask, Django, FastAPI, Microservices Architecture, RESTful APIs
Machine Learning Platform: PyTorch, TensorFlow, HuggingFace Ecosystem, MLOps, Model Deployment
NLP & AI Systems: Transformers, LangChain, NLTK, SpaCy, Custom Model Development
Data Engineering: PostgreSQL, MySQL, ChromaDB, FAISS, Vector Databases, ETL Pipelines
Cloud & DevOps: AWS (EC2, S3, Lambda), Docker, Kubernetes, CI/CD, Production Monitoring
Development Ecosystem: Git, Linux, Agile Methodologies, Code Review, Technical Leadership

PROFESSIONAL EXPERIENCE

AI Engineer June 2025 – Present
Nova Tech *Dubai, UAE (Remote)*

- Developed AI agent based cashflow management copilot system
- Architected intelligent trading bot using real time market data analysis with 85% accuracy with AI Agent
- Established company wide AI governance policy framework ensuring compliance with UAE financial regulations
- Led implementation of multi agent systems for portfolio management, risk assessment, and automated decision making

NLP Engineer August 2023 – Present
Proxima AI *Pakistan*

- Engineered chatbot platform achieving 50% reduction in response latency
- Built production-ready Speech-to-Text and Text-to-Speech systems for Urdu language processing
- Developed scalable image generation pipeline using Stable Diffusion, processing 10K+ requests daily
- Established technical mentorship program, resulting in 30% improvement in team delivery metrics
- Optimized deployment workflows reducing infrastructure costs by 25% while improving system reliability

Data Research Analyst January 2023 – June 2023
Technexia *Pakistan*

- Developed automated data intelligence systems using Python, processing 1M+ data points monthly
- Built scalable web scraping infrastructure with rate limiting and error handling mechanisms
- Created executive dashboards and analytical reports driving strategic business decisions

KEY ENGINEERING PROJECTS

TinyGemma-Urdu | *PyTorch, Transformer Architecture, Model Optimization* 2025

- Architected Google Gemma model from ground up with 0.96M parameters, achieving competitive performance
- Implemented custom transformer components including multi-head attention and positional encoding
- Optimized training pipeline with gradient checkpointing and mixed precision for efficient resource utilization

Whisper-Urdu-PyTorch | *Deep Learning, Audio Processing, Production Deployment* 2025

- Engineered complete Whisper architecture with 135M parameters for enterprise-grade Urdu speech recognition
- Built sophisticated encoder-decoder transformer with custom attention mechanisms and audio preprocessing
- Implemented real-time inference optimization achieving 3x faster processing with maintained accuracy

Urdu-BERT Foundation Model | *NLP Research, Custom Tokenization, MLOps* 2025

- Developed BERT model from scratch with custom byte-pair encoding tokenizer for Urdu text processing

- Implemented masked language modeling with distributed training across multiple GPUs
 - Established comprehensive evaluation framework with automated testing and performance benchmarking
- SmolUrxtral - Mixtral MoE** | *Advanced Architecture, Distributed Computing* 2025
- Designed Mixture of Experts architecture with 53M parameters and intelligent routing mechanisms
 - Implemented load balancing algorithms for efficient expert utilization and inference optimization
 - Achieved 40% reduction in computational overhead while maintaining model performance standards
- AI-Powered HR Management System** | *Conversational AI* 2025
- Developed comprehensive HR management AI chatbot handling all HR operations
 - Implemented intelligent agent for employee onboarding, performance evaluation, and policy management, taking leave request, making payrolls
 - Built natural language interface for leave requests, payroll queries, and benefits administration
 - Achieved 90% automation rate in routine HR tasks, reducing processing time by 75%
- Real-time Cricket Commentary System** | *Computer Vision, NLP, Audio Processing* 2025
- Engineered real-time cricket commentary system using LLM, TTS, and OpenCV for local cricket courts
 - Developed computer vision pipeline for ball tracking, player detection, and game state analysis
 - Integrated large language models for generating contextual commentary with cricket domain knowledge
 - Deployed edge computing solution processing video streams with sub-second latency for live commentary
- Intelligent Calling AI Agent** | *Voice AI, Natural Language Processing, Telephony* 2024
- Built sophisticated calling AI agent with natural conversation capabilities for customer service automation
 - Implemented voice recognition and synthesis pipeline with real-time speech processing
 - Designed context-aware dialogue management system handling complex multi-turn conversations
- Vikk AI - Legal Assistant** | *Full-Stack Development, Cloud Architecture* 2024
- Built scalable legal AI platform using microservices architecture on AWS with 99.9% uptime
 - Designed RESTful API with comprehensive user management system
 - Implemented advanced prompt engineering techniques resulting in 20% improvement in user satisfaction
- Urdu ASR & Voice Cloning** | *NVIDIA Nemo, XTTSv2, PyTorch, Data Processing* 2024
- Fine-tuned NVIDIA's Nemo ASR model for Urdu language with high transcription accuracy
 - Created custom Urdu dataset by transcribing YouTube audiobook and published on HuggingFace
 - Trained XTTSv2 model for realistic Urdu voice cloning with distinctive voice characteristics
 - Successfully replicated specific speaker's voice patterns for text-to-speech applications
- Intelligent Database Query System** | *NL2SQL, Database Optimization* 2023
- Developed natural language to SQL translation system with 95% query accuracy rate
 - Built secure database abstraction layer with query optimization and connection pooling
 - Implemented real-time query execution with comprehensive error handling and logging

EDUCATION

Bachelor of Science in Computer Science <i>University of Karachi</i>	September 2024 – August 2028 <i>Karachi, Pakistan</i>
Diploma in Computer Information Technology <i>Aligarh Institute of Technology</i>	October 2020 – October 2023 <i>Pakistan</i>

PROFESSIONAL CERTIFICATIONS & INDUSTRY RECOGNITION

Google Cloud - Advanced Encoder-Decoder Architecture & Transformer Models Specialization
 AWS - Practical Data Science on AWS Cloud Specialization with Production ML Focus
 IBM - Comprehensive Data Science Professional Certificate with Deep Learning Concentration
 Kaggle Notebook Expert - Top 1% contributor with 50K+ views on machine learning tutorials
 Distinguished Technical Speaker: "Advanced HuggingFace Transformers and LLMs" at SMIU Conference
 Open Source Maintainer: HuggingFace Organization "2Damnwav" with 10+ production-ready models
 Research Contributor: Multiple peer-reviewed publications in multilingual NLP and AI systems